Strict and weak strict stability imply local unique volume-constrained minimization with isolated conical singularities

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Allowed standard prerequisites (editorial).

- Classical research prerequisites cited by the original authors. Complete original bibliography retained; all source-local main estimates, constructions, supporting proofs and non-adjacent continuations retained.

1 Introduction

Isoperimetric boundaries minimise area for a fixed enclosed volume, with regularity theory ensuring smoothness away from a closed singular set of codimension seven. In [MN26] we showed that this result is sharp: in a given closed manifold, one can construct a variety of singular isoperimetric boundaries saturating this dimension bound on the singular set. Since isoperimetric boundaries can a priori possess any constant mean curvature (CMC), one might expect that 'most' isoperimetric boundaries should not be minimal. However, the constructions of [MN26] rested on a transfer of local area-minimisation to global isoperimetry, and hence the resulting singular isoperimetric boundaries all arose as minimal hypersurfaces. Motivated by this, we establish here the following result:

Theorem 1. *On a closed manifold of dimension at least eight, there exist singular isoperimetric boundaries with non-zero constant mean curvature with respect to some Riemannian metric.*

This shows the effectiveness of the results of [MNP25]; there it was shown that for a generic choice of Riemannian metric and enclosed volume pair (or for fixed enclosed volume and generic Riemannian metric) on a closed eight manifold, every isoperimetric region with respect to this pair (or metric) has smooth boundary. While the generic regularity results of [LW25] ensure that all minimal isoperimetric boundaries (e.g. as produced in both [MN26] and [Niu24]) are smooth in a closed eight manifold equipped with a generic Riemannian metric, this does not account for isolated singularities arising in isoperimetric boundaries with non-zero CMC (e.g. as produced by Theorem 1). Indeed, the isolated singularities in these hypersurfaces are only guaranteed to be perturbed away by the results of [MNP25], which account for any possible CMC value.

In [MN26, Corollary 2 & Remark 4] we showed that any smooth, connected, null-cobordant manifold could arise as an isoperimetric boundary with non-zero CMC value. This rested on the transfer of strict stability to volume-constrained minimisation available for smooth hypersurfaces by the results

of [Whi94] and [Gro96]; this is not known to hold generally in the presence of a singular set. Here, we establish this transfer for certain isolated singularities, generalising these results. This is the main technical result on which the constructions in Theorem 1 are based:

Theorem 2. *If $\Sigma = \partial\Omega$ is a strictly stable H-CMC hypersurface with isolated singularities modelled on strictly stable, strictly minimising regular tangent cones, then Ω uniquely minimises the natural area-type functional $\mathcal{F}_H$, and is consequently uniquely volume-constrained minimising, in a tubular neighbourhood of Σ. The conclusion of unique volume-constrained minimisation in a tubular neighbourhood also holds under the assumption that the CMC hypersurface is weakly strictly stable.*

For precise definitions of the natural area-type functional $\mathcal{F}_H$, strict stability of CMC hypersurfaces in the presence of isolated singularities, and the volume-constrained minimisation referred to above, see Subsection 1.2. As a consequence of Theorem 2, the isoperimetric boundaries constructed in Theorem 1 are unique for their enclosed volume; similar uniqueness holds for the examples produced in [MN26] and [Niu24] (see also the generic uniqueness results of [Niu26]). Finally, we note that the minimal (i.e. zero CMC value) case of Theorem 2 was established in [Sma99].

1.1 Strategy

In order to establish Theorem 1 from Theorem 2 we utilise an adaptation of the 'stretching' procedure introduced in [MN26, Theorem 1]. This reduces the problem to exhibiting connected, separating CMC hypersurfaces $\Sigma = \partial\Omega$ containing isolated singularities, such that Ω is volume-constrained minimising in a tubular neighbourhood of Σ.

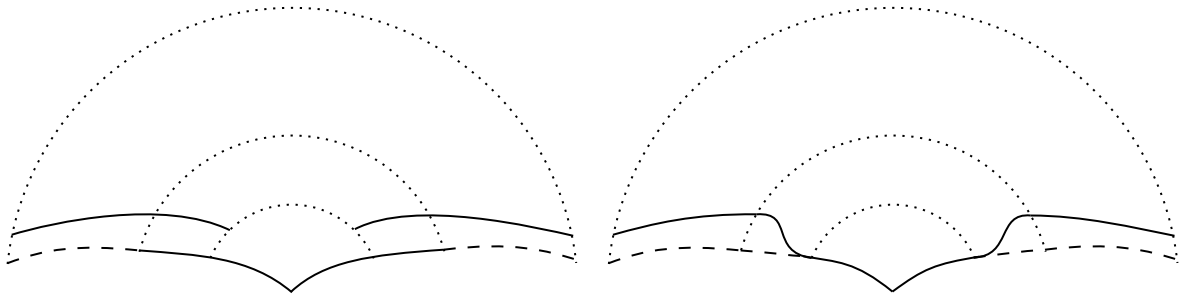


Figure 1: The graphic on the left above depicts smooth CMC graphs over a strictly stable minimal hypersurface (solid line outside the inner dotted ball) and a CMC hypersurface with an isolated singularity arising as a graph over a minimal hypercone (in the inner dotted ball). The solid line in the graphic on the right depicts a portion of Σ, obtained by interpolating (via gluing with a cutoff function) the two hypersurfaces (solid lines) depicted in the left graphic together in an annulus away from the singular point.

In order to construct Σ as described above, we utilise an adaptation of the constructions of [CHS84] (carried out in [Irv17]) which produce CMC hypersurfaces with isolated singularities as graphs over regular minimal hypercones. The hypersurface Σ is then constructed over strictly stable minimal hypersurfaces with isolated singularities as produced in [MN26, Theorem 3] (based on [Sma99] and [Sma00]): near each singular point Σ is the CMC graph produced by [CHS84] (where the metric is Euclidean), and away from the singular set Σ is comprised of CMC graphs (which exist by strict stability of the minimal hypersurface). These are then glued to the [CHS84] examples in annular regions around the isolated singularities. A conformal change in metric ensures the resulting hypersurface Σ is then a CMC hypersurface. This construction is depicted in Figure 1.

When the regular minimal hypercones in the construction of Σ above are further assumed to be strictly stable, we show that Σ is strictly stable (as defined in Subsection 1.2). By the above-mentioned reduction using the techniques of [MN26], to establish Theorem 1 it remains only to show that Ω is volume-constrained minimising in a tubular neighbourhood of Σ, which is Theorem 2.

We briefly recall how [Whi94] establishes the transfer of strict stability to volume-constrained minimisation in the smooth case. Consider a smooth H-CMC hypersurface $\Sigma = \partial\Omega$ and the natural area-type functional $\mathcal{F}_H(E) = \mathrm{Per}(E) - H \cdot \mathrm{Vol}(E)$ defined on Caccioppoli sets $E \subset M$. One then constructs a sequence of smooth barrier hypersurfaces ∂S_j (whose connected components converge to Σ smoothly as graphs as $j \to \infty$) and observes that, since Σ is smooth, for any sequence $\Sigma_j = \partial\Omega_j$ of $\mathcal{F}_H$-minimisers in S_j with $\Omega_j \Delta \Omega \subset S_j$, we have that $\Sigma_j \to \Sigma$ as $C^{1,\alpha}$ graphs as $j \to \infty$. By computing a Taylor expansion of the functional $\mathcal{F}_H$ for graphs over Σ, the strict stability assumption ensures that $\mathcal{F}_H(\Omega) < \mathcal{F}_H(\Omega_j)$ for sufficiently large $j \geq 1$ unless $\Sigma_j = \Sigma$; one thus deduces that Σ is uniquely $\mathcal{F}_H$-minimising (and hence volume-constrained minimising) in S_j for $j \geq 1$ sufficiently large.

To emulate this approach when Σ is a strictly stable H-CMC hypersurface containing isolated singularities modelled on strictly stable, strictly minimising regular tangent cones, we will require a control on $\mathcal{F}_H(\Omega_j)$ in balls around the singular points (where graphical convergence will fail). This control rests both on the construction of smooth barrier hypersurfaces ∂S_j (as above) and on a height-to-graphing radius bound for a (now potentially singular) sequence Σ_j which now only converges as $C^{1,\alpha}$ graphs to Σ on compact subsets away from the singular points. The smooth barrier hypersurfaces ∂S_j are constructed in a similar manner to those in [Sma00, Lemma 4] by exploiting the CMC analogue of [HS85] provided in [Les23] and [BL25]. We note that a similar height-to-graphing radius bound was deduced in the proof of [MNP25, Theorem 4] in order to ensure volume-control along sequences of convergent isoperimetric regions; a key difference here is that we deduce a super-quadratic (as opposed to super-linear) exponent under the stronger assumption of strict stability on the tangent cones. With this estimate in hand, we can excise the singular set and compute a Taylor expansion of $\mathcal{F}_H$ up to an error controlled by our height-to-graphing radius estimate. This error term can then be absorbed in the calculation, and the first statement of Theorem 2 then follows.

A similar line of reasoning to the above, now accounting for an imposed volume constraint and emulating the smooth case in [Gro96], also applies when assuming that Σ is only weakly strictly stable, establishing only the weaker conclusion of volume-constrained minimisation in this case; this gives the second statement of Theorem 2.

We now proceed as follows: Section 2 records both the adaptations of the constructions of [CHS84] to the CMC setting and results on the growth rates of CMC hypersurfaces near isolated singularities, before establishing Theorem 2. Section 3 constructs various examples of singular CMC hypersurfaces over singular minimal hypersurfaces and concludes by establishing Theorem 1.

1.2 Notation

We now collect some notation and definitions that will be used throughout this work:

- Let (M, g) be a smooth, closed (i.e. compact with empty boundary) Riemannian manifold of dimension $n + 1 \geq 8$. Without loss of generality we will assume that M is connected. Given a smooth Riemannian metric, g, on M we denote dist_g for the distance function on M, $B^g_r(p)$ the open geodesic ball in M of radius $r > 0$ centred at $p \in M$, $A^g(p; r, s)$ for the annulus of radii $0 \leq r < s$ centred at $p \in M$, $\exp^g_x$ for the exponential map based at $x \in M$, $\mathrm{Vol}_g(E)$ for the

volume of a measurable set $E \subset M$, and $\int_E u \, dV_g$ for the integral of an integrable function u, all with respect to the metric g. In $\mathbb{R}^{n+1}$ we write g_{eucl} for the Euclidean metric and omit metric dependence from our notation.

- By a **hypersurface**, $\Sigma \subset M$, we will mean a closed subset $\Sigma = \text{Reg}(\Sigma) \cup \text{Sing}(\Sigma)$, where the **regular set**, $\text{Reg}(\Sigma)$, is a smooth (or potentially $C^{1,\alpha} \cap W^{2,2}$ for some $\alpha \in (0,1)$) embedded manifold of dimension n, and the **singular set**, $\text{Sing}(\Sigma)$, is a (possibly empty) closed set of Hausdorff dimension at most $n-7$. We write $B_r^g(\text{Sing}(\Sigma))$ for the points $x \in M$ with $\text{dist}_g(x, \text{Sing}(\Sigma)) < r$ and $\text{graph}_\Sigma^g(u)$ the normal graph of u over Σ. Such hypersurfaces will arise as boundaries of Caccioppoli sets (defined below), which we may regard as integral n-currents with $\mathbb{Z}$ coefficients when the regular part is oriented, or n-chains with $\mathbb{Z}_2$ coefficients otherwise. We will also write $|\Sigma|_g$ for the multiplicity-one varifold associated to Σ and denote the varifold metric
$$\mathbf{F}(V,W) = \sup\{\|V\|(f) - \|W\|(f) \,|\, f \in C^1(M), |f| \leq 1, |Df| \leq 1\},$$
defined for integer rectifiable varifolds V where $\|V\|$ denotes the weight measure of V.

- A measurable set $E \subset M$ is a **Caccioppoli set** if the indicator function of E is of bounded variation, or equivalently if it has finite **perimeter** with respect to the metric g so that
$$\text{Per}_g(E) = \sup\left\{ \int_E \text{div}_g X \, dV_g \;\middle|\; X \in \Gamma(TM), g(X,X) \leq 1 \right\} < \infty,$$
where div_g is the divergence with respect to the metric g and $\Gamma(TM)$ is the set of vector fields on M; we write $\text{Per}_g(E;U)$ for the perimeter relative to a measurable set $U \subset M$ by restricting the integral above to U. We write $\mathcal{C}(S)$ for the set of Caccioppoli sets in a set S, and note that $\mathcal{C}(S)$ is independent of the choice of metric on S. For a Caccioppoli set, $U \in \mathcal{C}(M)$, we denote by $[U]$ the associated integral $(n+1)$-current with coefficients in $\mathbb{Z}_2$; then $\partial U = \text{Spt}(\partial[U])$.

- We say that $\Omega \in \mathcal{C}(M)$ is an **isoperimetric region** of enclosed volume $t \in (0, \text{Vol}_g(M))$ if
$$\text{Per}_g(\Omega) = \inf_{E \in \mathcal{C}(M)} \{\text{Per}_g(E) \,|\, \text{Vol}_g(E) = t\}.$$
The existence of isoperimetric regions of a given enclosed volume in M is then guaranteed by the direct method of the calculus of variations (e.g. see [Mor16, Section 13.2]). Moreover, we say that $\Omega \in \mathcal{C}(M)$ is a **volume-constrained minimiser** in an open set $U \subset M$ if
$$\text{Per}_g(\Omega; U) = \inf_{E \in \mathcal{C}(M)} \{\text{Per}_g(E;U) \,|\, \Omega \Delta E \subset\subset U \text{ and } \text{Vol}_g(\Omega \cap U) = \text{Vol}_g(E \cap U)\}.$$

- Given an isoperimetric region Ω as above, we denote by $\Sigma = \partial\Omega$ the closed, two-sided, embedded CMC hypersurface; by [GMT83] and [Mor03], we ensure that $\text{Sing}(\Sigma)$ is of Hausdorff dimension at most $n-7$. We denote by ν_g the outward-pointing unit normal to Σ with respect to the metric g.

- We often consider the area-type functional
$$\mathcal{F}_H(E) = \text{Per}_g(E) - H \cdot \text{Vol}_g(E),$$
defined on $E \in \mathcal{C}(M)$ with respect to g. Critical points of $\mathcal{F}_H$ have boundaries with distributional mean curvature equal to H, and minimisers of $\mathcal{F}_H$ are volume-constrained minimising in M.

- Suppose that $\mathbf{C} \subset \mathbb{R}^{n+1}$ is a **regular** minimal hypercone, so that $\mathrm{Sing}(\mathbf{C}) \subset \{0\}$. We then denote the smooth, closed hypersurface $S = \mathbf{C} \cap \mathbb{S}^n \subset \mathbb{R}^{n+1}$ as its **link**. By parameterising $\mathbf{C}$ in radial coordinates, $(r, \omega) \in (0, \infty) \times S$, we decompose (as in [Sim68]) the Jacobi operator, $L_{\mathbf{C}}$, of $\mathbf{C}$ as

$$L_{\mathbf{C}} = \partial_r^2 + \left(\frac{n-1}{r}\right)\partial_r + \frac{1}{r^2}(\Delta_S + |\mathrm{I\!I}_S|^2),$$

where $\mathrm{I\!I}_S$ is the second fundamental form of S in $\mathbb{S}^n$. We let $\mu_1 < \mu_2 \leq \mu_3 \leq \ldots \nearrow +\infty$ be the eigenvalues of $-(\Delta_S + |\mathrm{I\!I}_S|^2)$, and $\varphi_1, \varphi_2, \ldots$ be the corresponding $L^2(S)$-orthonormal eigenfunctions where $\varphi_1 > 0$. By [Sim68], $\mathbf{C}$ is **stable** if and only if $\mu_1 \geq -\frac{(n-2)^2}{4}$ and we say (as in [CHS84]) that $\mathbf{C}$ is **strictly stable** if $\mu_1 > -\frac{(n-2)^2}{4}$; the relationships above regarding inequalities for μ_1 and the stability of $\mathbf{C}$ are made explicit in [CHS84, Section 4]. Moreover, we say (as in [HS85]) that a minimal hypercone, $\mathbf{C}$, as above is **strictly minimising** if there is a $\Theta > 0$ such that for each $\varepsilon > 0$, whenever T is an integer rectifiable current with $\mathrm{Spt}(T) \subset \mathbb{R}^{n+1} \setminus B_\varepsilon(0)$ and $\partial T = \partial(\mathbf{C} \cap B_1(0))$ we have

$$|\mathbf{C} \cap B_1(0)| \leq |T| - \Theta\varepsilon^n.$$

For $\Lambda > 0$ we denote

$$\mathscr{C}_\Lambda = \{\text{stable regular minimal hypercones, } \mathbf{C} \subset \mathbb{R}^{n+1}, \text{ with } \|\mathbf{C}\|(B_1(0)) \leq \Lambda\}$$

and write $\mathscr{C}$ for the stable regular minimal hypercones without density bound.

- By [CHS84], a general **Jacobi field**, $v \in C^\infty_{\mathrm{loc}}(\mathbf{C} \setminus \{0\})$, on $\mathbf{C}$ (i.e. a solution to $L_{\mathbf{C}} v = 0$) is given by a linear combination of homogeneous Jacobi fields

$$v(r, \omega) = \sum_{j \geq 1} (v_j^+(r) + v_j^-(r))\varphi_j(\omega),$$

where for each $j \geq 1$ we write

$$v_j^+(r) = c_j^+ \cdot r^{\gamma_j^+} \qquad \text{and} \qquad v_j^-(r) = \begin{cases} c_j^- \cdot r^{\gamma_j^-}, & \text{if } \mu_j > -\frac{(n-2)^2}{4}, \\ c_j^- \cdot r^{\gamma_j^-} \log(r) & \text{if } \mu_j = -\frac{(n-2)^2}{4}, \end{cases}$$

for some constants $c_j^\pm \in \mathbb{R}$, and

$$\gamma_j^\pm = \gamma_j^\pm(\mathbf{C}) = -\frac{n-2}{2} \pm \sqrt{\mu_j + \frac{(n-2)^2}{4}}.$$

We call the collection of exponents, $\gamma_j^\pm$ for $j \geq 1$, the **asymptotic spectrum** of $\mathbf{C}$, which we denote by $\Gamma(\mathbf{C})$. We observe that if we choose $\gamma \in (-\frac{n-2}{2}, \gamma_1^+(\mathbf{C}))$, then whenever $\mathbf{C}$ is strictly stable we have that

$$\frac{n}{1-\gamma} > 2. \tag{1}$$

- Following [Sma99] we define the weighted Sobolev space $W_0^{1,2,-2}(\Sigma)$ to be the completion of $C_c^1(\mathrm{Reg}(\Sigma))$ with respect to the norm

$$\|u\|^2_{W^{1,2,-2}(\Sigma)} = \int_\Sigma (|\nabla u|^2 + u^2\rho^{-2})\, dA_g,$$

where $\rho : M \to [0,\infty)$ is the distance to $\mathrm{Sing}(\Sigma)$. We then say that Σ is **strictly stable** if Σ is a singular CMC hypersurface and there is $\lambda > 0$ such that

$$\delta^2 \mathrm{Area}_\Sigma(u) = \int_\Sigma (|\nabla u|^2 - (|\mathrm{I\!I}_\Sigma|^2 + \mathrm{Ric}_g(\nu,\nu))u^2)\, dA_g \geq \lambda \int_\Sigma u^2 \rho^{-2}\, dA_g \tag{2}$$

for all $u \in W_0^{1,2,-2}(\Sigma)$; if (2) holds only for those $W_0^{1,2,-2}$ functions with zero integral, we say that Σ is **weakly strictly stable**. We note that strict stability (and weak strict stability) with respect to the ρ^{-2}-weighted L^2 and $W^{1,2,-2}$ norms are equivalent up to a change in the choice of λ. Indeed, using the fact that $|\mathrm{I\!I}_\Sigma|^2 + \mathrm{Ric}_g(\nu,\nu) \leq C\rho^{-2}$ we have

$$\int_\Sigma |\nabla u|^2\, dA_g = \delta^2 \mathrm{Area}_\Sigma(u) + \int_\Sigma (|\mathrm{I\!I}_\Sigma|^2 + \mathrm{Ric}_g(\nu,\nu))u^2\, dA_g \leq \left(1 + \frac{C}{\lambda}\right) \delta^2 \mathrm{Area}_\Sigma(u),$$

and so

$$\|u\|^2_{W^{1,2,-2}(\Sigma)} = \int_\Sigma |\nabla u|^2\, dA_g + \int_\Sigma u^2 \rho^{-2}\, dA_g \leq \left(1 + \frac{C+1}{\lambda}\right) \delta^2 \mathrm{Area}_\Sigma(u),$$

so (2) can be phrased with the squared $W^{1,2,-2}$ norm of u up to replacing λ with $\frac{\lambda}{\lambda+C+1}$.

- We let $\mathrm{inj}(x; M, g)$ denote the injectivity radius at $x \in M$ with respect to the metric g and for each hypersurface $\Sigma = \partial\Omega$ we define the **regularity scale** $r_S(x)$ depending on M, g, and Σ, at a point $x \in \mathrm{Reg}(\Sigma)$ to be the supremum among all $r \in (0, \mathrm{inj}(x; M, g)/2)$ such that both of the following properties hold:
  - $r^2 \|\mathrm{Rm}_g\|_{C^0(B^g_r(x))} + r^3 \|\nabla \mathrm{Rm}_g\|_{C^0(B^g_r(x))} \leq 1/10$, where Rm_g denotes the Riemann curvature tensor of M with respect to the metric g.
  - After pulling back by $\exp^g_x$ to T_xM we have
  $$\frac{1}{r}(\exp^g_x)^{-1}(\Sigma) \cap B_1(0) = \mathrm{graph}^{g_{\mathrm{eucl}}}_L(u) \cap B_1(0),$$
  for some hyperplane $L \subset T_xM$ and $u \in C^3(L)$ with $\|u\|_{C^3(L)} \leq 1/10$.
- For $\phi \in C^k_{\mathrm{loc}}(\Sigma)$ with $k = 0, 1, 2$ we define the following norm on $E \subset \Sigma$ to be
$$\|\phi\|_{C^k_*(E)} = \sup_{x \in E} \sum_{j=0}^{k} r_S(x)^{j-1} |\nabla^j \phi(x)|,$$
which is invariant under the scaling of ϕ to $\lambda\phi$ and g to $\lambda^2 g$ for $\lambda > 0$.
- By [Sim83a, Theorem 5], for any $p \in \mathrm{Sing}(\Sigma)$, we may express Σ in **conical coordinates** near p over its tangent cone $\mathbf{C}_p\Sigma$; precisely, for any $\varepsilon > 0$, there exist $\phi \in C^2(\mathbf{C}_p\Sigma \setminus \{0\})$ and $r_p(\varepsilon) > 0$ such that $\|\phi\|_{C^2_*((\mathbf{C}_p \cap B_{r_p(\varepsilon)}(0)) \setminus \{0\})} < \varepsilon$ and
$$\mathrm{graph}^{g_{\mathrm{eucl}}}_{\mathbf{C}_p\Sigma}(\phi) \cap B_{r_p(\varepsilon)}(0) = (\exp^g_p)^{-1}\left(\Sigma \cap B^g_{r_p(\varepsilon)}(p)\right).$$

2 Local behaviour

In this section we recall the constructions of [CHS84] (and the extension due to [Irv17]) which produce prescribed mean curvature hypersurfaces with isolated singularities modelled on regular minimal hypercones. We then record the definition of growth rates for such hypersurfaces (c.f. [MNP25] and [LW25]) and establish a growth rate monotonicity for local isoperimetric regions with respect to barriers. With this growth rate monotonicity in hand, we conclude the section by establishing both statements in Theorem 2.

2.1 Constructions over regular minimal hypercones

We first recall the construction of [CHS84] which produces singular minimal hypersurfaces modelled on regular minimal hypercones, explaining how modifications of their techniques produce prescribed mean curvature hypersurfaces; this extension was carried out explicitly in [Irv17].

In [CHS84] examples of (embedded) minimal hypersurfaces in $\mathbb{R}^{n+1}$ with isolated singularities at the origin are constructed by perturbing the link, $\mathbf{C} \cap \mathbb{S}^n$, of a given regular minimal hypercone $\mathbf{C}$ and solving $\mathcal{M}_{\mathbf{C}} u = 0$ where $\mathcal{M}_{\mathbf{C}}$ is the mean curvature operator of graphs of functions on $\mathbf{C}$. When the functions u has appropriate decay towards the origin, the resulting graph

$$\mathrm{graph}_{\mathbf{C}}(u) = \{x + u(x)\nu(x) \,|\, x \in \mathbf{C} \setminus \{0\}\},$$

where ν is an outward unit normal to $\mathbf{C}$, is embedded. Minimal hypersurfaces are constructed by first finding (by separation of the radial variable) solutions of the inhomogeneous Jacobi field equation with appropriately fast decay to zero on approach to the origin, applying Schauder estimates for small data in this linear problem, and utilising an appropriate fixed-point theorem to find the desired solutions of the non-linear problem. The same fixed-point arguments work for small enough inhomogeneous data f and provide solutions to the prescribed mean curvature equation $\mathcal{M}_{\mathbf{C}}(u) = f$ (shown explicitly in [Irv17]). More precisely, the results of [CHS84] and [Irv17] show:

Theorem 3 ([CHS84],[Irv17])**.** *For $n \geq 3$ and $\alpha \in (0,1)$, if $\mathbf{C} \subset \mathbb{R}^{n+1}$ is a regular minimal hypercone, then there exist constants $\delta \in [0,1) \setminus \Gamma(\mathbf{C})$ depending on n and $\mathbf{C}$, and $\varepsilon \in (0,1), C > 0$ depending on n, α, and $\mathbf{C}$ such that the following holds. For all $f \in C^{0,\alpha}(\mathbf{C} \cap B_1(0) \setminus \{0\})$ with*

$$\sup_{0<r<\frac{1}{2}} \|f(r\,\cdot)\|_{C^{0,\alpha}(\mathbf{C}\cap A(0;1,2))} \leq \varepsilon,$$

there exists $u \in C^{2,\alpha}(\mathbf{C} \cap B_1(0) \setminus \{0\})$, with $\mathrm{graph}_{\mathbf{C}}(u)$ an embedded hypersurface, solving

$$\mathcal{M}_{\mathbf{C}} u = f \quad \text{in } \mathbf{C} \cap B_1(0) \setminus \{0\},$$

and satisfying the decay condition

$$\sup_{0<r<\frac{1}{2}} r^{-2+\delta} \|u(r\,\cdot)\|_{C^{2,\alpha}(\mathbf{C}\cap A(0;\frac{1}{2},1))} < C \sup_{0<r<\frac{1}{2}} \|f(r\,\cdot)\|_{C^{0,\alpha}(\mathbf{C}\cap A(0;\frac{1}{2},1))}.$$

Moreover, when f is constant and $\mathbf{C}$ is strictly stable, then, after potentially choosing $\varepsilon \in (0,1)$ smaller, $\mathrm{graph}_{\mathbf{C}}(u)$ is strictly stable in the sense of (2) in $B_1(0)$.

2.2 Growth rates over regular minimal hypercones

We now sharpen results in [MNP25, Appendix A] concerning the growth rates near isolated singular points of converging sequences of hypersurfaces that will be utilised in proving Theorem 2. We first recall some definitions:

Definition 1. *Given a hypersurface $\Sigma \subset M$, $p \in \mathrm{Sing}(\Sigma)$ isolated, $v \in L^2_{\mathrm{loc}}(\Sigma \setminus \{p\})$, $\mathbf{C} \in \mathscr{C}$, $w \in C^2_{\mathrm{loc}}(\mathbf{C} \setminus \{0\})$, $r > 0$, and $K > 1$, we define:*

- *The **growth rate** of v at p, Given $\gamma \in \mathbb{R}$, to be*

$$J^{\gamma}_{K;\Sigma,g}(v;r) = \left(\int_{A(p;K^{-1}r,r)} v^2 \cdot \rho^{-n-2\gamma} \, dA_g \right)^{\frac{1}{2}},$$

  where $\rho(x) = \mathrm{dist}_g(x,p)$.

- *The **asymptotic rate** of w at infinity to be*

$$\mathcal{AR}_\infty(w) = \inf\left\{\gamma \,\middle|\, \lim_{s\nearrow+\infty} \int_{A(0;s,2s)} w^2(x)\cdot|x|^{-n-2\gamma}\, d\|\mathbf{C}\|(x) = 0\right\},$$

*and the **growth rate** of w on $\mathbf{C}$, given $\gamma\in\mathbb{R}$, to be*

$$J^\gamma_{K;\mathbf{C}}(w;r) = \left(\int_{A(0;K^{-1}r,r)} w^2(x)\cdot|x|^{-n-2\gamma}\, d\|\mathbf{C}\|(x)\right)^{\frac{1}{2}}.$$

- *For a stationary integral varifold V in $\mathbb{R}^{n+1}\setminus B_{R_0}$ for some $R_0>0$, we say that V is **asymptotic to $\mathbf{C}$ at infinity** if there is $w\in C^2_{\mathrm{loc}}(\mathbf{C}\setminus\{0\})$ with*

$$V\llcorner(\mathbb{R}^{n+1}\setminus B_{R_1}) = |\mathrm{graph}^{g_{\mathrm{eucl}}}_{\mathbf{C}\setminus B_{R_1}}(w)|\llcorner(\mathbb{R}^{n+1}\setminus B_{R_1})$$

for some $R_1\geq R_0$ and $\|w\|_{C^2_*(A(0;R,2R))}\to 0$ as $R\to\infty$; for such a varifold V asymptotic to $\mathbf{C}$ at infinity we write $\mathcal{AR}_\infty(V)=\mathcal{AR}_\infty(w)$.

For the above definitions we adopt the convention that $\sup\emptyset=-\infty$ and $\sup\mathbb{R}=+\infty$.

Remark 1. *By expressing Σ in conical coordinates over its tangent cone $\mathbf{C}_p\Sigma$, there is some constant $C>0$, depending on the radius in which these coordinates are defined, such that*

$$C^{-1}J^\gamma_{K;\mathbf{C}}(u;r)^2 \leq J^\gamma_{K;\Sigma,g}(u;r)^2 \leq CJ^\gamma_{K;\mathbf{C}}(u;r)^2.$$

We will utilise the following monotonicity property for the growth rate, whose statement and proof are an adaptation of the dichotomy appearing in [LW25, Lemma F.5]:

Lemma 1. *For $\sigma>0$, $\Lambda>\frac{n-2}{2}$, and $K>2$ (chosen based on σ,Λ as in [LW25, Corollary 6.2]), there exist constants $\delta,C>0$, depending on σ,Λ, and K, for which the following holds. Consider a strictly minimising $\mathbf{C}\in\mathscr{C}_\Lambda$ and $\gamma\in(-\frac{n-2}{2},\gamma_1^+(\mathbf{C}))$ such that*

$$\mathrm{dist}_{\mathbb{R}}\left(\gamma,\Gamma(\mathbf{C})\cup\{-(n-2)/2\}\right)\geq\sigma.$$

Let g be a Riemannian metric on $B_2(0)$ so that in normal coordinates we have $\|g-g_{\mathrm{eucl}}\|_{C^4}\leq\delta$. Suppose that $\Omega\in\mathcal{C}(B_2(0))$, with $\Sigma=\partial\Omega$ an H-CMC hypersurface with respect to g such that $\mathrm{Sing}(\Sigma)\cap B_2(0)=\{0\}$ and

$$|\Sigma|\llcorner B_2(0) = |\mathrm{graph}_{\mathbf{C}}(w)|\llcorner B_2(0) \text{ with } \|w\|_{C^2_*(\mathbf{C}\cap B_2(0))}<\delta.$$

Suppose that $\{\Omega_j\}_{j\geq1}\subset\mathcal{C}(B_2(0))$, with $\Sigma_j=\partial\Omega_j$ a $C^{1,\alpha}\cap W^{2,2}$ hypersurface with distributional mean curvature $H_j\in L^\infty(\Sigma_j)$ with respect to g for each $j\geq1$, such that $\|H_j-H\|_{L^\infty(\Sigma_j\cap B_2(0))}\to0$ and $\mathbf{F}(|\Sigma|\llcorner B_2(0),|\Sigma_j|\llcorner B_2(0))\to0$ as $j\to\infty$. Then, if we define for each $j\geq1$ the graphing radii τ_j and functions v_j to be

$$\tau_j = \inf\{s\in(0,1)\mid |\Sigma_j|\llcorner A(0;s,1) = |\mathrm{graph}_\Sigma(v_j)|\llcorner A(0;s,1) \text{ and } \|v_j\|_{C^1_*(\Sigma\cap A(0;s,1))}\leq\delta\} \tag{3}$$

we have for sufficiently large $j\geq1$ and $t\in(2K^3\tau_j,1)$ that

$$J^\gamma_{K;\Sigma,g}(v_j;K^{-1}t)\leq\max\{J^\gamma_{K;\Sigma,g}(v_j;t),C\|H_j-H\|_{L^\infty(\Sigma_j\cap B_2(0))}\}. \tag{4}$$

Proof. We first note that in the case where Σ_j has mean curvature H (i.e. $H_j = H$) for all $j \geq 1$, the proof reduces exactly to the statement of [LW25, Lemma F.5] since the equation for the difference of the graphs (see (5)) has right-hand side zero. The alternative in the dichotomy there is then ruled out by the strict minimisation assumption on $\mathbf{C}$ by [Wan20, Theorem 4.5(1)] (explained in full for the general case below).

The general case also follows by adapting the proof of [LW25, Lemma F.5] and appealing to [LW25, Corollary 6.2]. With the notation there we take $g_j^0 = g_j^1 = g$, Φ for w, and set $w_j = v_j \circ \Phi$ (the height of Σ_j over Σ pulled back to $\mathbf{C}$). Let $\varepsilon, K > 0$ be as in [LW25, Corollary 6.2], and choose $\delta > 0$ sufficiently small so that, by [LW25, Theorem B.1 and Remark 6.1], the error terms $\vec{B}_0, B_1$ below satisfy [LW25, (F.14)] with ε replaced by $\varepsilon/2$. Applying [LW25, Theorem B.1] we see that on $\mathbf{C} \cap A(0; 2\tau_j, 1)$ the functions w_j weakly solve

$$\mathrm{div}_{\mathbf{C}}(\nabla_{\mathbf{C}} w_j + \vec{B}_0(x)) + |\mathrm{I\!I}_{\mathbf{C}}|^2 w_j + |x|^{-1} B_1(x) = \widetilde{H}_j, \tag{5}$$

where $\vec{B}_0, B_1$ satisfy the estimate [LW25, (F.14)] with $\frac{\varepsilon}{2}$. The right-hand side of (5) above is initially $(H_j \circ \Phi_j - H)F + H(F - F')$, where F and F' are the area densities of Σ_j and Σ respectively and Φ_j is the graph of Σ_j over $\mathbf{C}$ as in [LW25, Theorem B.1]. However, since $|H| \leq C_0\delta$ for some constant $C_0 > 0$ (since $\|g - g_{\mathrm{eucl}}\|_{C^4} \leq \delta$ and $\|w\|_{C^2_*(\mathbf{C} \cap B_2(0))} < \delta$), the second term is absorbed into B_1, and so we may set $\widetilde{H}_j = (H_j \circ \Phi_j - H)F$ with

$$\|\widetilde{H}_j\|_{L^\infty(\mathbf{C} \cap A(0;2\tau_j,1))} \leq C_1 \|H_j - H\|_{L^\infty(\Sigma_j \cap B_2(0))}$$

for a constant $C_1 > 0$ (into which we absorb the constant in Remark 1). We now set $C_2 = \frac{\varepsilon}{2} K^{-n}$ and $C = C_1 C_2^{-1}$; this latter C is the constant we will obtain for the statement.

For each $t \in (2K^3\tau_j, 1)$ the rescalings $\hat{w}_j(x) = \frac{w_j(tx)}{t}$ weakly solve (5) on $\mathbf{C} \cap A(0; K^{-3}, 1)$ where now $\|\widetilde{H}_j\|_{L^\infty(\mathbf{C} \cap A(0;K^{-3},1))} \leq tC_1 \|H_j - H\|_{L^\infty(\Sigma_j \cap B_2(0))}$. Thus, if (4) fails at this t then we have

$$\begin{aligned}\|\widetilde{H}_j\|_{L^\infty(\mathbf{C} \cap A(0;K^{-3},1))} &\leq tC_1 \|H_j - H\|_{L^\infty(\Sigma_j \cap B_2(0))} \leq tC_2 J^\gamma_{K;\mathbf{C}}(w_j; K^{-1}t) \\ &\leq C_2 t^{\gamma-1} J^\gamma_{K;\mathbf{C}}(w_j; K^{-1}t) \leq C_2 J^\gamma_{K;\mathbf{C}}(\hat{w}_j; K^{-1}) \leq \frac{\varepsilon}{2} \|\hat{w}_j\|_{L^2(A(0;K^{-3},1))},\end{aligned}$$

using Remark 1, $t \leq 1$, $\gamma < 0$, and $|x|^{-n-2\gamma} \leq K^{2n}$ on $A(0; K^{-2}, K^{-1})$. Absorbing $\widetilde{H}_j$ into B_1 we see that $\hat{w}_j$ satisfies [LW25, (33)–(34)] and so iterating [LW25, Corollary 6.2] gives [LW25, (F.15)] with the $\kappa\|g^1 - g^0\|_{C^4}$ term appearing there replaced by $C\|H_j - H\|_{L^\infty(\Sigma_j \cap B_2(0))}$.

Then, if (4) fails for infinitely many $j \geq 1$ the same reasoning following [LW25, (F.15)] shows that $\tau_j > 0$ for sufficiently large $j \geq 1$ and hence $\tau_j^{-1}|\Sigma_j|$ converges to a stationary integral varifold V_∞ which differs from $\mathbf{C}$ but is asymptotic to it at infinity with $\mathcal{AR}_\infty(V_\infty) < \gamma$. However, this conclusion is precluded by the choice of $\gamma \in (-\frac{n-2}{2}, \gamma_1^+(\mathbf{C}))$ since the assumption that $\mathbf{C}$ is strictly minimising ensures $\mathcal{AR}_\infty(V_\infty) \geq \gamma_1^+$ by [Wan20, Theorem 4.5(1)]. Thus the desired statement holds. □

Remark 2. *We note that the above lemma differs from the dichotomy stated in [MNP25, Appendix A] in the choice of γ, the fact that we do not consider changing metrics, and that we only work with the C^1_* norm. This is permitted since the weak formulations used in the proof require only $W^{1,2}_{\mathrm{loc}}$ functions.*

2.3 Strict stability to volume-constrained minimisation

Throughout this subsection we will assume $\Sigma = \partial\Omega$ is a connected, separating H-CMC hypersurface with finitely many isolated singularities modelled on strictly stable, strictly minimising regular tangent cones of multiplicity one; the results of [Sim83a] show that such tangent cones are unique. Under the assumption of strict stability we will show that Ω is $\mathcal{F}_H$-minimising in a tubular neighbourhood of Σ. Moreover, under the assumption of weak strict stability we show Ω is volume-constrained minimising in a tubular neighbourhood of Σ. Combining both statements yields Theorem 2.

We first construct a sequence of barriers for Σ, which do not require any global stability assumptions. More precisely, for each $j \geq 1$ we construct nested sets $A_j \subset \Omega \subset B_j$ converging to Ω in the L^1 sense as $j \to \infty$. Moreover, denoting $S_j = B_j \setminus A_j$, we show that each ∂S_j (disjoint from Σ) is a pair of smooth hypersurfaces converging smoothly as graphs to Σ away from each singular point as $j \to \infty$.

As shown in [Les23, Theorem 7.3] (c.f. [HS85, Theorem 4.1]), the assumption of strict stability and strict minimisation of the regular tangent cone at each singular point ensures that Σ is in fact locally $\mathcal{F}_H$-minimising around each singular point. Fix $r > 0$ sufficiently small so that the balls $B^g_{8r}(p)$ are pairwise disjoint for each $p \in \mathrm{Sing}(\Sigma)$, so that normal coordinates over $\Sigma \setminus \{\rho < r\}$ are defined, and sufficiently small for the application of [Les23, Theorem 8.7] that follows below.

By [Les23, Theorem 8.7] (which records the adaptation of [BL25, Theorem 2] to the Riemannian setting), for a sequence $t_j \to 0$ as $j \to \infty$ there are $\mathcal{F}_H$-minimisers $R_p^{\pm t_j}$ in $B^g_{8r}(p)$ with boundaries equal to the $\pm t_j$-normal graphs over $\Sigma \cap \partial B^g_{8r}(p)$. We set $S_p^{\pm t_j} = \partial(R_p^{\pm t_j}) \llcorner B^g_{8r}(p)$, so that each $S_p^{\pm t_j}$ is an entirely smooth H-CMC hypersurface lying on one side of Σ (we take the convention that $R_p^{-t_j} \subset \Omega \cap B^g_{8r}(p) \subset R_p^{+t_j}$). Moreover, we have that $S_p^{\pm t_j} \to \Sigma \cap B^g_{8r}(p)$ in Hausdorff distance as $t_j \to 0$, and thus on the annulus $r \leq \rho \leq 7r$ we can write each hypersurface as a smooth normal graph $w_p^{\pm t_j}$ over Σ by Allard's theorem (e.g. as stated in [Sim83b, Chapter 5]) whenever t_j is sufficiently small. By the Harnack inequality (which we can apply since both Σ and $S_p^{\pm t_j}$ are H-CMC) it then follows that for some $c > 0$ we have both

$$c^{-1}|t_j| \leq |w_p^{\pm t_j}| \leq c|t_j| \quad \text{and} \quad \max_{2r \leq \rho \leq 6r} |\nabla^k w_p^{\pm t_j}| \leq c|t_j| \quad \text{for} \quad k = 1, 2.$$

Also by the choice of $r > 0$ above, for $|t|$ small we can define the smooth parallel hypersurfaces

$$C_t = \{\exp_x(t\nu(x)) \,|\, x \in \Sigma,\ \rho(x) \geq r\},$$

where $\nu(x)$ is the outward-pointing unit normal to Σ at a point $x \in \mathrm{Reg}(\Sigma)$. Since Σ is an H-CMC hypersurface we compute directly that the mean curvature H_{C_t} of the smooth hypersurfaces C_t satisfies

$$|H_{C_t} - H| \leq C|t| \quad \text{on } \{\rho \geq r\},$$

for some constant $C > 0$ depending on r and Σ.

We now glue the above hypersurfaces to form barriers. For this, we choose a smooth increasing cutoff function $\eta : [0, \infty) \to \mathbb{R}$, with $\eta = 0$ on $[0, 3r]$ and $\eta = 1$ on $[5r, \infty)$, and on each annulus $\{2r \leq \rho \leq 6r\}$ around $p \in \mathrm{Sing}(\Sigma)$ we set

$$u_p^{\pm t_j} = (1 - \eta)w_p^{\pm t_j} \pm \eta t_j;$$

thus the graph of $u_p^{\pm t_j}$ interpolates between $C_{\pm t_j}$ and $S_p^{\pm t_j}$. We denote the resulting hypersurfaces by $\Gamma_j^{\pm}$ so that S_j is defined implicitly to be the region with $\partial S_j = \Gamma_j^+ \cup \Gamma_j^-$ containing Σ. More concretely, we define

$$\Gamma_j^{\pm} = \bigcup_{p \in \mathrm{Sing}(\Sigma)} (S_p^{\pm t_j} \cap B^g_{3r}(p)) \cup \mathrm{graph}(u_p^{\pm t_j}) \cup (C_{\pm t_j} \cap (M \setminus B^g_{5r}(p)))$$

which are then smooth hypersurfaces.

Finally, for each $j \geq 1$ we write $M = A_j \cup S_j \cup B_j^c$ so that $S_j = B_j \setminus A_j$ is chosen with outward-pointing unit normal to Γ_j^+ pointing into B_j^c; by construction we see that then for each $j \geq 1$ we have $A_j \subsetneq \Omega \subsetneq B_j$ with $|A_j \Delta \Omega|, |B_j \Delta \Omega|, |A_j \Delta B_j| \to 0$ as $j \to \infty$. Note that ∂S_j then has two connected components $\Gamma_j^{\pm}$ with $\partial A_j = \Gamma_j^-$, $\partial B_j = \Gamma_j^+$, and $\partial S_j = \partial A_j \cup \partial B_j$ for each $j \geq 1$.

We will also make use of the following Taylor expansions in the proof of Theorem 2:

Remark 3. *(Taylor expansions) Let $\Sigma = \partial\Omega$ be an H-CMC hypersurface with finitely many isolated singularities with strictly stable, strictly minimising regular tangent cones of multiplicity one as above. For $\psi \in C^1_*(\Sigma)$ denote by Ω_ψ the region obtained from Ω with boundary given by*

$$\Sigma_\psi = \{\exp_x^g(\psi(x)\nu(x)) \mid x \in \Sigma\}.$$

We consider the Taylor expansions of the perimeter and volume. Following the computations in [LW25, Appendix B], we see that there exist constants $\varepsilon_0, C > 0$, depending on Σ and (M, g) such that for $\|\psi\|_{C^1_*(\Sigma)} \leq \varepsilon_0$ we have

$$\mathrm{Per}_g(\Omega_\psi) - \mathrm{Per}_g(\Omega) = H\int_\Sigma \psi\, dA_g + \frac{1}{2}\int_\Sigma \left(|\nabla_\Sigma \psi|^2 - (|\mathrm{II}_\Sigma|^2 + \mathrm{Ric}_g(\nu,\nu) - H^2)\psi^2\right) dA_g + R_P(\psi),$$

and

$$\mathrm{Vol}_g(\Omega_\psi) - \mathrm{Vol}_g(\Omega) = \int_\Sigma \psi\, dA_g + \frac{H}{2}\int_\Sigma \psi^2\, dA_g + R_V(\psi).$$

where

$$|R_P(\psi)| + |R_V(\psi)| \leq C\|\psi\|_{C^1_*(\Sigma)} \cdot \|\psi\|^2_{W^{1,2,-2}(\Sigma)}.$$

Hence, writing $Q_L(\psi,\psi) = \int_\Sigma (|\nabla_\Sigma\psi|^2 - (|\mathrm{II}_\Sigma|^2 + \mathrm{Ric}_g(\nu,\nu))\psi^2)\, dA_g$, we have

$$\mathcal{F}_H(\Omega_\psi) - \mathcal{F}_H(\Omega) = \frac{1}{2}Q_L(\psi,\psi) + R_F(\psi), \tag{6}$$

where

$$|R_F(\psi)| = |R_P(\psi) - HR_V(\psi)| \leq C\|\psi\|_{C^1_*(\Sigma)} \cdot \|\psi\|^2_{W^{1,2,-2}(\Sigma)}. \tag{7}$$

We now establish the first statement in Theorem 2, which follows from:

Theorem 4. *If Σ is strictly stable, then Ω is uniquely locally $\mathcal{F}_H$-minimising in a tubular neighbourhood of Σ. In particular, Ω is volume-constrained minimising in this tubular neighbourhood.*

Proof. We will show that Ω is uniquely $\mathcal{F}_H$-minimising in the sets S_j constructed above for sufficiently large $j \geq 1$; both statements then follow since $\mathcal{F}_H$-minimisers are volume-constrained minimisers (seen by considering fixed-volume competitors). To this end, for each $j \geq 1$ we let $\Omega_j \in \mathcal{C}(M)$ solve

$$\mathcal{F}_H(\Omega_j) = \inf_{E \in \mathcal{C}(M)} \{\mathcal{F}_H(E) \mid E\Delta\Omega \subset S_j\};$$

we will show that $\Omega_j = \Omega$ for $j \geq 1$ sufficiently large. Note first that by regularity results for the isoperimetric problem with respect to an obstacle (e.g. see [Vol10, Theorem 1.1] for a formulation in Riemannian manifolds) and since Ω_j is volume-constrained minimising (since it is $\mathcal{F}_H$-minimising) in S_j and ∂S_j is a union of smooth hypersurfaces, we have for some $\alpha \in (0,1)$ that $\Sigma_j = \partial\Omega_j$ is a $C^{1,\alpha} \cap W^{2,2}$ hypersurface away from a closed set of Hausdorff dimension at most $n-7$ with bounded

distributional mean curvature. Moreover, since $\Gamma_j^{\pm}$ are smooth H-CMC hypersurfaces in the balls $B^g_{2r}(p)$ for each $p \in \mathrm{Sing}(\Sigma)$, the mean curvature estimates in [Vol10, Theorem 1.1] ensure that Σ_j is a (potentially singular) H-CMC hypersurface in each $B^g_{2r}(p)$ for $j \geq 1$ sufficiently large.

Since $\Omega_j \Delta \Omega \subset S_j$ for each $j \geq 1$, by their construction we have $A_j \subset \Omega_j \subset B_j$ and thus as both $|A_j \Delta \Omega|, |B_j \Delta \Omega| \to 0$ we have $|\Omega_j \Delta \Omega| \to 0$ as $j \to \infty$ also. Noting that the $\mathcal{F}_H$-minimisation of Ω_j gives

$$\mathrm{Per}(\Omega_j) - H|\Omega_j| \leq \mathrm{Per}(\Omega) - H|\Omega|,$$

the fact that $|\Omega_j \Delta \Omega| \to 0$ as $j \to \infty$ combined with the lower semicontinuity of perimeter gives $\mathrm{Per}(\Omega_j) \to \mathrm{Per}(\Omega)$ as $j \to \infty$. Thus, by [DT13, Proposition A.1], we see that $\mathbf{F}(|\Sigma|, |\Sigma_j|) \to 0$ as $j \to \infty$ and hence $\Sigma_j \to \Sigma$ as $C^{1,\alpha}$ graphs away from singular points of Σ by Allard's theorem (e.g. as stated in [Sim83b, Chapter 5]) as $j \to \infty$.

For each $p \in \mathrm{Sing}(\Sigma)$ fix some $\gamma_p \in (-\frac{n-2}{2}, \gamma_1^+(\mathbf{C}_p))$ a distance $\sigma > 0$ away from the asymptotic spectrum, and choose $\gamma \in (-\frac{n-2}{2}, \min_{p \in \mathrm{Sing}(\Sigma)}(\gamma_p))$. For this choice of $\sigma > 0$ and some $\Lambda > \frac{n-2}{2}$ we then apply Lemma 1 to obtain $\delta > 0$ for which its conclusions hold. Using conical coordinates for Σ around each $p \in \mathrm{Sing}(\Sigma)$ (as defined in Subsection 1.2) we can write the graphing radii for Σ_j over Σ around each $p \in \mathrm{Sing}(\Sigma)$ as

$$s_j = \max_{p \in \mathrm{Sing}(\Sigma)} \inf\{\tau > 0 \,|\, \|v_j\|_{C^1_*(A^g(p;\tau,1))} \leq \delta\},$$

this ensures for $j \geq 1$ sufficiently large that

$$\Sigma_j \setminus B^g_{s_j}(p) = \mathrm{graph}^g_{\Sigma}(v_j) \setminus B^g_{s_j}(p).$$

Moreover, by the strict stability and strict minimisation assumption on $\mathbf{C}_p$ for each $p \in \mathrm{Sing}(\Sigma)$, we have that both (1) holds for γ and (4) holds over each $\mathbf{C}_p$.

By defining $a_j = \|v_j\|_{L^2(\Sigma \setminus B^g_{K^{-2}}(\mathrm{Sing}(\Sigma)))}$ we will now show, as in the proof of [MNP25, Theorem 4], that

$$s_j \leq a_j^{\frac{1}{1-\gamma}} = a_j^{\frac{2+\beta}{n}} \tag{8}$$

where γ is chosen as above and $\beta > 0$ (the second equality coming from (1)). Indeed, if this fails then by defining $\hat{v}_j(x) = \frac{v_j(s_j x)}{s_j}$ on $A(0; 1, \frac{1}{K s_j})$ we see that $J^{\gamma_p}_{K;\mathbf{C}_p}\left(\hat{v}_j; \frac{1}{K s_j}\right) \to 0$ since, by writing $y = s_j x$, we compute

$$J^{\gamma_p}_{K;\mathbf{C}_p}\left(\hat{v}_j; \frac{1}{K s_j}\right)^2 = s_j^{-2+2\gamma_p} \int_{A\left(0; \frac{1}{K^2}, \frac{1}{K}\right)} v_j(y)^2 |y|^{-n-2\gamma_p} \, d\|\mathbf{C}_p\|(y) \leq K^{2n+4\gamma_p} s_j^{-2+2\gamma_p} a_j^2,$$

so that under the assumption that $a_j \leq s_j^{1-\gamma}$ we ensure that

$$J^{\gamma_p}_{K;\mathbf{C}_p}\left(\hat{v}_j; \frac{1}{K s_j}\right) \leq C s_j^{(\gamma_p - \gamma)} \to 0.$$

However, by writing (4) for $\hat{v}_j$ (with $H_j = H$ for all $j \geq 1$ since the Σ_j are H-CMC in $B^g_{2r}(p)$ for each $p \in \mathrm{Sing}(\Sigma)$) we deduce that $J^{\gamma_p}_{K;\mathbf{C}_p}(\hat{v}_j; K^l)$ is increasing for integers $3 \leq l \leq \log_K(\frac{1}{s_j}) - 1$ and hence we have

$$J^{\gamma_p}_{K;\mathbf{C}_p}\left(\hat{v}_j; K^3\right) \leq J^{\gamma_p}_{K;\mathbf{C}_p}\left(\hat{v}_j; \frac{1}{K s_j}\right).$$

Since $\hat{v}_j \to h \in C^2_{\mathrm{loc}}(\mathbf{C}_p \setminus B_1(0))$ in C^2_{loc} where h is the graph of a stationary varifold, V, which is asymptotic to but not equal to $\mathbf{C}_p$ (obtained as a limit of $s_j^{-1}|\Sigma_j|$ by the minimality of s_j), for all large j we see that

$$J^{\gamma_p}_{K;\mathbf{C}_p}\left(\hat{v}_j; K^3\right) > \frac{1}{2} J^{\gamma_p}_{K;\mathbf{C}_p}\left(h; K^3\right) > 0$$

where the right-hand term is strictly positive since $h \neq 0$ (since $V \neq \mathbf{C}_p$). This is a contradiction since we saw that $J^{\gamma_p}_{K;\mathbf{C}_p}\left(\hat{v}_j; \frac{1}{Ks_j}\right) \to 0$ as $j \to \infty$; hence (8) holds as desired.

Next, we modify (in a similar manner to Figure 1) the sets Ω_j inside of $B^g_{2s_j}(\mathrm{Sing}(\Sigma))$ in order to construct sets $\widetilde{\Omega}_j$ agreeing with Ω in $B^g_{s_j}(\mathrm{Sing}(\Sigma))$. To this end, we choose a smooth increasing cutoff function $\zeta \in C^\infty([0,\infty))$ with $\zeta = 0$ on $[0,1]$ and $\zeta = 1$ on $[2,\infty)$. For $j \geq 1$ and $x \in B^g_{2s_j}(p) \cap \Sigma$ for each $p \in \mathrm{Sing}(\Sigma)$ we define

$$\widetilde{v}_j(x) = \zeta\left(\frac{\mathrm{dist}_g(x,p)}{s_j}\right) v_j(x) \tag{9}$$

for $x \in A^g(p; s_j, 2s_j)$ while setting $\widetilde{v}_j = v_j$ on $\Sigma \setminus B^g_{2s_j}(p)$ and $\widetilde{v}_j = 0$ on $B^g_{s_j}(p)$ for each $p \in \mathrm{Sing}(\Sigma)$. Since for each $j \geq 1$ we have $\|v_j\|_{C^1_*(A^g(p;s_j,1))} \leq \delta$ for each $p \in \mathrm{Sing}(\Sigma)$, we ensure $\|\widetilde{v}_j\|_{C^1_*(A^g(p;s_j,1))} \leq C\delta$ for a constant $C > 0$ depending on ζ. Moreover, we note that $a_j \leq \|\widetilde{v}_j\|_{L^2(\Sigma)}$ for each $j \geq 1$ by construction.

For each $j \geq 1$ we then denote by $\widetilde{\Omega}_j$ the Caccioppoli set whose boundary is given by the global normal graph $\widetilde{\Sigma}_j = \mathrm{graph}^g_\Sigma(\widetilde{v}_j)$, with $\widetilde{\Omega}_j \Delta \Omega_j \subset S_j$. The calculations in Remark 3 then provide an analogous computation to that of [Whi94, (1) in the proof of Theorem 2] and show that

$$\mathcal{F}_H(\widetilde{\Omega}_j) - \mathcal{F}_H(\Omega) = \frac{1}{2} Q_L(\widetilde{v}_j, \widetilde{v}_j) + R_F(\widetilde{v}_j) \geq \frac{\lambda}{2}\|\widetilde{v}_j\|^2_{W^{1,2,-2}(\Sigma)} - C\|\widetilde{v}_j\|_{C^1_*(\Sigma)}\|\widetilde{v}_j\|^2_{W^{1,2,-2}(\Sigma)}$$

where we used (2) from the strict stability (with constant $\lambda > 0$) of Σ with respect to the $W^{1,2,-2}$ norm (as in Subsection 1.2) and (7) to bound R_F.

Noting that $\widetilde{\Omega}_j$ differs from Ω_j only in $B^g_{2s_j}(\mathrm{Sing}(\Sigma))$, we see that

$$|\mathcal{F}_H(\widetilde{\Omega}_j) - \mathcal{F}_H(\Omega_j)| \leq C s_j^n,$$

for a constant $C > 0$ depending on Σ and (M, g), and thus for $j \geq 1$ sufficiently large we have

$$\mathcal{F}_H(\Omega_j) - \mathcal{F}_H(\Omega) \geq \frac{\lambda}{2}\|\widetilde{v}_j\|^2_{W^{1,2,-2}(\Sigma)} - \|\widetilde{v}_j\|_{C^1_*(\Sigma)}\|\widetilde{v}_j\|^2_{W^{1,2,-2}(\Sigma)} - C s_j^n.$$

Combining $a_j \leq \|\widetilde{v}_j\|_{L^2(\Sigma)}$ with (1) and (8) we see that $s_j^n \leq \|\widetilde{v}_j\|^{2+\beta}_{L^2(\Sigma)}$ for some $\beta > 0$. Moreover, if we choose $\delta < \frac{\lambda}{8C}$ above so that $\|\widetilde{v}_j\|_{C^1_*(\Sigma)} < \frac{\lambda}{8}$ (noting that Lemma 1 holds for all sufficiently small $\delta > 0$) and use the fact that $\|\widetilde{v}_j\|_{L^2(\Sigma)} \to 0$ we see that for $j \geq 1$ sufficiently large we have

$$\mathcal{F}_H(\Omega_j) - \mathcal{F}_H(\Omega) \geq \frac{\lambda}{4}\|\widetilde{v}_j\|^2_{W^{1,2,-2}(\Sigma)}.$$

The above inequality ensures that, if $\Omega_j \neq \Omega$ we must have

$$\mathcal{F}_H(\Omega_j) > \mathcal{F}_H(\Omega)$$

whenever $j \geq 1$ is sufficiently large. This contradicts the assumption that the Ω_j were $\mathcal{F}_H$-minimising in S_j unless $\Omega_j = \Omega$. Thus, Ω is uniquely $\mathcal{F}_H$-minimising in S_j. Since each S_j contains a tubular neighbourhood of Σ by their construction, we see that the desired statement holds. □

Under the assumption of weak strict stability we also obtain the following statement, which combined with Theorem 4 yields the full statement of Theorem 2:

Theorem 5. *If Σ is weakly strictly stable, then Ω is uniquely locally volume-constrained minimising in a tubular neighbourhood.*

Proof. We emulate the approach in the smooth case carried out in [Gro96], showing that weak strict stability of Σ implies strict stability for a modified functional (obtained by adding a quadratic term to $\mathcal{F}_H$ that does not affect the first variation). We then show that Ω uniquely minimises this modified functional, and hence is volume-constrained minimising, in S_j for all sufficiently large $j \geq 1$. Since the approach is similar to that of Theorem 4 above, we only outline the necessary modifications.

For $C > 0$ sufficiently large we consider the modified functional

$$\mathcal{E}_C(E) = \mathcal{F}_H(E) + \frac{C}{2}\big(\mathrm{Vol}_g(E) - \mathrm{Vol}_g(\Omega)\big)^2$$

defined on $E \in \mathcal{C}(M)$, and will show the strict stability of Σ with respect to $\mathcal{E}_C$ (c.f. [Gro96, Lemma 5]). The compactness argument used in the proof of [Gro96, Lemma 5] does not apply directly here, since $W_0^{1,2,-2}(\Sigma)$ does not compactly embed into the space of ρ^{-2}-weighted $L^2(\Sigma)$ functions. Instead, write $u = w + (\int_\Sigma u\, dA_g)\,\psi_0$ for a fixed $\psi_0 \in C_c^\infty(\mathrm{Reg}(\Sigma))$ with $\int_\Sigma \psi_0\, dA_g = 1$, so that $\int_\Sigma w\, dA_g = 0$. We then see from the weak strict stability inequality (2) applied to w, the boundedness of Q_L on $W_0^{1,2,-2}(\Sigma)$, and Young's inequality, that for $C > 0$ sufficiently large we have

$$\delta^2\mathcal{E}_C(u) = Q_L(u,u) + C\left(\int_\Sigma u\, dA_g\right)^2 \geq \frac{\lambda}{4}\|u\|^2_{W^{1,2,-2}(\Sigma)}$$

for all $u \in W_0^{1,2,-2}(\Sigma)$. In other words, Σ is strictly stable for $\mathcal{E}_C$ with constant $\frac{\lambda}{4}$ when $C > 0$ is sufficiently large.

We now show that Ω is uniquely $\mathcal{E}_C$-minimising, and hence volume-constrained minimising, in the sets S_j constructed above for sufficiently large $j \geq 1$. To this end, for each $j \geq 1$ we let $\Omega_j \in \mathcal{C}(M)$ solve

$$\mathcal{E}_C(\Omega_j) = \inf_{E \in \mathcal{C}(M)} \{\mathcal{E}_C(E) \,|\, E\Delta\Omega \subset S_j\};$$

we will show that $\Omega_j = \Omega$ for $j \geq 1$ sufficiently large. Just as in the proof of Theorem 4 above, by regularity results for the isoperimetric problem with respect to an obstacle (e.g. see [Vol10, Theorem 1.1] for a formulation in Riemannian manifolds) and since Ω_j is volume-constrained minimising (for its volume) in S_j and ∂S_j is a union of smooth hypersurfaces, we have for some $\alpha \in (0,1)$ that $\Sigma_j = \partial\Omega_j$ is a $C^{1,\alpha} \cap W^{2,2}$ hypersurface away from a closed set of Hausdorff dimension at most $n-7$ with bounded distributional mean curvature. Exactly as in the proof of Theorem 4 above, since $|\Omega_j \Delta \Omega| \to 0$ as $j \to \infty$, the lower semicontinuity of the perimeter, [DT13, Proposition A.1], and Allard's theorem, we see that $\Sigma_j \to \Sigma$ as $C^{1,\alpha}$ graphs away from singular points of Σ as $j \to \infty$.

We now apply Lemma 1 to the Σ_j in conical coordinates for Σ near each $p \in \mathrm{Sing}(\Sigma)$. We first note that, since the first variations of $\mathcal{F}_H$ and $\mathcal{E}_C$ differ by a volume difference term, the distributional mean curvature of Σ_j satisfies

$$|H_j - H| = |C(\mathrm{Vol}_g(\Omega_j) - \mathrm{Vol}_g(\Omega))| = d_j$$

wherever it is defined (e.g. see [Vol10, Theorem 1.1]). This means that the mean curvature difference in (4) will not necessarily be zero near $\mathrm{Sing}(\Sigma)$ (unlike in the proof of Theorem 4).

Fix, as in the proof of Theorem 4, $\gamma_p \in (-\frac{n-2}{2}, \gamma_1^+(\mathbf{C}_p))$ a distance $\sigma > 0$ from the asymptotic spectrum and $\gamma \in (-\frac{n-2}{2}, \min_{p\in\mathrm{Sing}(\Sigma)} \gamma_p)$. For this choice of $\sigma > 0$ and some $\Lambda > \frac{n-2}{2}$ we then apply Lemma 1 to obtain $\delta, \widetilde{C} > 0$ (writing C for the constant $\widetilde{C}$ there) for which its conclusions hold. Using conical coordinates for Σ around each $p \in \mathrm{Sing}(\Sigma)$ (as defined in Subsection 1.2) we can write the graphing radii for Σ_j over Σ around each $p \in \mathrm{Sing}(\Sigma)$ as

$$s_j = \max_{p\in\mathrm{Sing}(\Sigma)} \inf\{\tau > 0 \,|\, \|v_j\|_{C^1_*(A^g(p;\tau,1))} \leq \delta\},$$

this ensures for $j \geq 1$ sufficiently large that

$$\Sigma_j \setminus B^g_{s_j}(p) = \mathrm{graph}^g_\Sigma(v_j) \setminus B^g_{s_j}(p).$$

Moreover, by the strict stability and strict minimisation assumption on $\mathbf{C}_p$ for each $p \in \mathrm{Sing}(\Sigma)$, we have that both (1) holds for γ and (4) holds over each $\mathbf{C}_p$.

We now aim to show that an analogue of (8) holds (up to a multiplicative constant) where we define $a_j = \|v_j\|_{L^2(\Sigma\setminus B^g_{K^{-2}\tilde{r}}(\mathrm{Sing}(\Sigma)))}$ for some sufficiently small $0 < \tilde{r} \leq r$ to be chosen. Let us set $b_j = a_j + \widetilde{C} d_j$ for each $j \geq 1$ and note that running the same argument (obtaining a contradiction by applying (4)) as in the proof of Theorem 4, we see that $s_j \leq b_j^{\frac{1}{1-\gamma}}$. It thus remains to show that $d_j \leq C_1 a_j$ since then we have $s_j^{1-\gamma} \leq b_j = a_j + \widetilde{C} d_j \leq (1 + \widetilde{C} C_1) a_j$, yielding the desired analogue of (8).

On $M \setminus B^g_{\tilde{r}}(\mathrm{Sing}(\Sigma))$ we have that $\Sigma_j = \mathrm{graph}^g_\Sigma(v_j)$ and hence we see that

$$|(\Omega_j \Delta \Omega) \setminus B^g_{\tilde{r}}(\mathrm{Sing}(\Sigma))| \leq 2 \int_{\Sigma\setminus B^g_{\tilde{r}}(\mathrm{Sing}(\Sigma))} |v_j| \, dA_g \leq C_0 a_j. \tag{10}$$

On each annular region $A^g(p; K^{-1}t, t)$ for $p \in \mathrm{Sing}(\Sigma)$, for $t = K^{-m}\tilde{r}$ for integers $m = 0, \ldots, M_t$ where M_t is the largest integer with $K^{-M_t}\tilde{r} \geq K^3 s_j$, we see that by Cauchy–Schwarz (and the fact that $n + 2\gamma_p > 0$ by assumption) we have

$$\int_{A^g(p;K^{-1}t,t)} |v_j| \, dA_g \leq \left(\int_{A^g(p;K^{-1}t,t)} \rho^{n+2\gamma_p} \, dA_g \right)^{\frac{1}{2}} J^{\gamma_p}_{K;\Sigma,g}(v_j; t) \leq C t^{n+\gamma_p} J^{\gamma_p}_{K;\Sigma,g}(v_j; t).$$

Using (4) repeatedly and summing over annuli, for fixed $j \geq 1$, we see that

$$\begin{aligned} |(\Omega_j \Delta \Omega) \cap A^g(p; K^3 s_j, \tilde{r})| &\leq C \sum_{m=0}^{M_j} \int_{\Sigma\cap A^g(p;t_m/K,t_m)} |v_j| \, dA_g \leq C \sum_{m=0}^{M_j} t_m^{n+\gamma_p} J_p(v_j; t_m) \\ &\leq C \max\{J_p(v_j; \tilde{r}), \tilde{C} d_j\} \sum_{m=0}^{M_j} t_m^{n+\gamma_p} \leq C_{\tilde{r}} a_j + c\widetilde{C} \tilde{r}^{n+\gamma_p} d_j. \end{aligned} \tag{11}$$

In $B^g_{2K^3 s_j}(\mathrm{Sing}(\Sigma))$ we have that

$$\begin{aligned} |(\Omega_j \Delta \Omega) \cap B^g_{2K^3 s_j}(\mathrm{Sing}(\Sigma))| &\leq C_2 s_j^{n+1} \leq C_2 s_j^{n+\gamma} s_j^{1-\gamma} \leq C_2 s_j^{n+\gamma} b_j \\ &= C_2 s_j^{n+\gamma}(a_j + \widetilde{C} d_j) < C_2 a_j + C_2 \widetilde{C} s_j^{n+\gamma} d_j. \end{aligned} \tag{12}$$

With the bounds (10), (11), and (12) above, and choosing an initial $0 < \tilde{r} \leq r$ sufficiently small based on the constants, we can absorb the terms involving d_j whenever $j \geq 1$ is sufficiently large (since $s_j \to 0$ and $n + \gamma > 0$) and obtain the bound $d_j \leq C_1 a_j$ as desired.

With the above estimate in hand, we modify the graphs exactly as in (9), obtaining sets $\widetilde{\Omega}_j$ for each $j \geq 1$ with $\partial\widetilde{\Omega}_j = \widetilde{\Sigma}_j = \mathrm{graph}^g_\Sigma(\widetilde{v}_j)$ and $\widetilde{\Omega}_j \Delta \Omega_j \subset S_j$. We then proceed using Remark 3 to see that since $\widetilde{\Omega}_j$ differs from Ω_j only in $B^g_{2s_j}(\mathrm{Sing}(\Sigma))$ we have

$$\mathcal{E}_C(\Omega_j) - \mathcal{E}_C(\Omega) \geq \frac{\lambda}{8}\|\widetilde{v}_j\|^2_{W^{1,2,-2}(\Sigma)} - \|\widetilde{v}_j\|_{C^1_*(\Sigma)}\|\widetilde{v}_j\|^2_{W^{1,2,-2}(\Sigma)} - Cs_j^n$$

for some $C > 0$ depending on Σ and (M, g). With the estimate $s_j \leq C_1 a_j^{\frac{1}{1-\gamma}}$ in hand, since $a_j \leq \|\widetilde{v}_j\|_{L^2(\Sigma)}$ we see that $s_j^n \leq \|\widetilde{v}_j\|^{2+\beta}_{L^2(\Sigma)}$ for some $\beta > 0$ (using (1)). Also, choosing $\delta < \frac{\lambda}{32C}$ above and using the fact that $\|\widetilde{v}_j\|_{L^2(\Sigma)} \to 0$ we see that for $j \geq 1$ sufficiently large we have

$$\mathcal{E}_C(\Omega_j) - \mathcal{E}_C(\Omega) \geq \frac{\lambda}{16}\|\widetilde{v}_j\|^2_{W^{1,2,-2}(\Sigma)}.$$

Just as in the conclusion of the proof of Theorem 4, we thus see that $\Omega_j = \Omega$ for sufficiently large $j \geq 1$ and hence Ω is uniquely $\mathcal{E}_C$-minimising in S_j. In particular, Ω is volume-constrained minimising in a tubular neighbourhood of Σ as desired. □

Remark 4. *In [MR10] it was shown that, in ambient dimensions at most seven, weakly strictly stable smooth CMC hypersurfaces are uniquely homologically area minimising for fixed volumes in L^1-neighbourhoods. The dimensional restriction stems from their reliance on [MR10, Proposition 1]. Though we do not pursue it here, it would be interesting to know if our Theorem 5 could be utilised to extend their result to allow for isolated singularities of the type considered here.*

3 Global behaviour

In this section we construct CMC hypersurfaces with prescribed singular sets by exploiting Theorem 3 in a similar manner to [MN26, Theorem 3] and [Sma00, Theorems 1 & 2]; in particular, we remove the assumption that the closed manifolds comprising the singular sets there need be oriented. We then use Theorem 2 and an adaptation of the arguments in [MN26, Corollary 2] to construct, under the assumption that all tangent cones are regular, strictly stable and strictly minimising, examples of singular isoperimetric boundaries with non-zeero CMC, thus establishing Theorem 1.

3.1 Constructions over singular minimal hypersurfaces

In order to find strictly stable CMC hypersurfaces with isolated singularities in any closed manifold of dimension at least eight, we use Theorem 3 to modify the constructions of [MN26, Theorem 3] and [Sma00, Theorem 1]:

Proposition 1. *Suppose that M is a closed manifold of dimension $n+1 \geq 8$, $L \geq 1$ is an integer, and $\{\mathbf{C}_i\}_{i=1}^L$ is a collection of strictly stable, non-planar regular minimal hypercones $\mathbf{C}_i \subset \mathbb{R}^{n+1}$. Then, for each $H \neq 0$ sufficiently small, there is a Riemannian metric g on M and an open set $\Omega \subset M$ with $\Sigma = \partial\Omega$ a strictly stable H-CMC hypersurface with L isolated singularities. Moreover, the unique tangent cone at each isolated singularity $p_i \in \mathrm{Sing}(\Sigma)$ for $1 \leq i \leq L$ is $\mathbf{C}_i$.*

Proof. Using the construction in the proofs of [MN26, Theorem 3] (taking each Γ_i there as a point) and [Sma00, Theorem 1] (noting that one does not require orientability for M), and utilising [Sma00, Proposition 2], we find an open set $\Omega_0 \subset M$ and a separating hypersurface $\Sigma_0 = \partial\Omega_0$ which is minimal and strictly stable with respect to some Riemannian metric g_0 on M. Moreover, $\mathrm{Sing}(\Sigma_0)$ consists of L

isolated points with unique tangent cones given by the $\{\mathbf{C}_i\}_{i=1}^L$, and g_0 is Euclidean in $B_1^{g_0}(\mathrm{Sing}(\Sigma_0))$ with $\Sigma_0 \cap B_1(p_i) = \mathbf{C}_i \cap B_2(p_i)$ for each $1 \leq i \leq L$. We note that we do not require the strict minimisation of the $\{\mathbf{C}_i\}_{i=1}^L$ here since we make no claim about the local area-minimisation of Σ_0.

By the strict stability of Σ_0, for all $H \neq 0$ sufficiently small we can invert the Jacobi operator of Σ_0 away from $B_{1/2}^{g_0}(\mathrm{Sing}(\Sigma_0))$ to find H-CMC hypersurfaces with boundary (with zero Dirichlet boundary data on $\Sigma_0 \cap \partial B_{1/2}^{g_0}(\mathrm{Sing}(\Sigma_0))$) as graphs v_H over $\Sigma_0 \setminus B_{1/2}^{g_0}(\mathrm{Sing}(\Sigma_0))$ (c.f. [MN26, Remark 4]).

Applying Theorem 3 with $f = H$ over each $\mathbf{C}_i$ we produce $u_i \in C^{2,\alpha}(\mathbf{C}_i \cap B_1(p_i) \setminus \{p_i\})$ with $\mathrm{graph}_{\mathbf{C}_i}(u_i)$ an embedded strictly stable H-CMC hypersurface in $B_1(p_i)$. For each $H \in \mathbb{R} \setminus \{0\}$ sufficiently small as above we construct a hypersurface which agrees with the graphs of the u_i near $\mathrm{Sing}(\Sigma_0)$ and after conformal metric perturbation will be shown to be a strictly stable H-CMC hypersurface.

We choose a smooth increasing cutoff function $\eta : [0, \infty) \to \mathbb{R}$ with $\eta = 0$ on $[0, 5/8]$ and $\eta = 1$ on $[7/8, \infty)$, and on $B_1^{g_0}(\mathrm{Sing}(\Sigma_0))$ we set

$$w_i = (1 - \eta)u_i + \eta v_H,$$

so that the graph of w_i interpolates between the graphs of u_i and v_H. We denote the resulting hypersurface by Σ, which is H-CMC away from $\cup_{p_i \in \mathrm{Sing}(\Sigma_0)} A^{g_0}(p_i; 5/8, 7/8)$, and the open set it bounds by Ω. This is the construction depicted in Figure 1.

By performing a conformal metric perturbation supported near the annuli $A^{g_0}(p_i; 5/8, 7/8)$ (c.f. [BM25, Proposition 1] and [Mar24, Proposition 2.1]) we produce a Riemannian metric g on M for which Σ is a strictly stable H-CMC hypersurface. The follows for sufficiently small non-zero H since the metric changes are localised to annuli around $\mathrm{Sing}(\Sigma_0)$ and can be made arbitrarily small. □

Remark 5. *It follows from the proof above that $g = g_0$ outside of $B_1^{g_0}(\mathrm{Sing}(\Sigma))$ with both metrics Euclidean in $B_{1/2}^{g_0}(\mathrm{Sing}(\Sigma))$. Moreover, one can ensure that the resulting metric is arbitrarily close to the original metric on M in $C^{k,\alpha}$ norm for each $k \geq 1$ and $\alpha \in (0, 1)$.*

If one is willing to forgo prescribing the ambient manifold, we can use a similar construction to the above to prescribe arbitrary unions of closed manifolds (not necessarily orientable) as the singular set of a strictly stable H-CMC hypersurface for any $H \in \mathbb{R}$:

Proposition 2. *Let $n + 1 \geq 8$, $H \in \mathbb{R}$, and for an integer $L \geq 1$ consider collections of closed Riemannian manifolds $\{(\Gamma_i, h_i)\}_{i=1}^L$ of dimension k_i, $\{(X_i, g_i)\}_{i=1}^L$ of dimension $n_i + 1$ with $k_i + n_i = n$, and a collection $\{\mathbf{C}_i\}_{i=1}^L$ of strictly stable, non-planar regular minimal hypercones of dimension n_i. Then, there is a closed $(n + 1)$-dimensional Riemannian manifold (M, g), formed as the connected sum $M = \#_{i=1}^L (X_i \times \Gamma_i)$ equipped with some Riemannian metric g, and an open set $\Omega \subset M$ such that $\Sigma = \partial\Omega \subset (M, g)$ is a strictly stable H-CMC hypersurface with*

$$\mathrm{Sing}(\Sigma) = \bigcup_{i=1}^{L} \Lambda_i,$$

where each Λ_i is an embedded copy of Γ_i. Moreover, for some $\sigma > 0$ sufficiently small, in a neighbourhood of each Λ_i, (M, g) is locally isometric to $B_\sigma^{n_i+1}(0) \times \Gamma_i$ endowed with the product metric $g_{\mathrm{eucl}} \times h_i$, with Σ in this region equal to $(\mathrm{graph}_{\mathbf{C}_i}(u_i) \cap B_\sigma^{n_i+1}(0)) \times \Gamma_i$ where the u_i are as in Theorem 3.

Proof. The construction in the proof of [Sma00, Theorem 2] provides the ambient manifold M along with a Riemannian metric g_0 and a separating strictly stable minimal hypersurface $\Sigma_0 = \partial\Omega_0 \subset (M, g_0)$ satisfying each of the desired conclusions save for the mean curvature condition. In particular, near each Λ_i we have that Σ_0 equal to $\mathbf{C}_i \times \Gamma_i$.

As in the proof of Proposition 1 we use Theorem 3 to produce H-CMC hypersurfaces as the graphs of functions $u_i \in C^{2,\alpha}(\mathbf{C}_i \cap B_1(p_i) \setminus \{p_i\})$. We then glue these graphs to Σ_0 in annular regions with a smooth cutoff and conformally perturb the metric (once to get H-CMC, then to get stability) to produce the strictly stable H-CMC hypersurface Σ as desired. □

Remark 6. *Since Theorem 3 allows one to prescribe non-constant mean curvature functions, in principle an adaptation of the above construction yields the existence of prescribed mean curvature hypersurfaces with arbitrary closed manifolds as their singular set; we did not pursue this.*

3.2 Isoperimetry of singular CMC hypersurfaces

The following result implies Theorem 1:

Theorem 6. *Suppose M is a closed manifold of dimension $n+1 \geq 8$, $L \geq 1$ is an integer, and $\{\mathbf{C}_i\}_{i=1}^L$ is a collection of strictly stable, strictly minimising, non-planar regular minimal hypercones $\mathbf{C}_i \subset \mathbb{R}^{n+1}$. Then, for each $H \neq 0$ sufficiently small there is a Riemannian metric h on M and a separating, strictly stable H-CMC hypersurface $\Sigma \subset (M, h)$ with L isolated singularities which bounds an isoperimetric region. Moreover, the unique tangent cone at each isolated singularity $p_i \in \mathrm{Sing}(\Sigma)$ for $1 \leq i \leq L$ is $\mathbf{C}_i$.*

Proof. The proof of Proposition 1 produces a separating, strictly stable minimal hypersurface $\Sigma_0 = \partial\Omega_0$ for some Riemannian metric g_0. Since the $\{\mathbf{C}_i\}_{i=1}^L$ are assumed to be strictly minimising we see that, as noted in the conclusion of the proof of [MN26, Theorem 3] (c.f. [Sma00, Pages 2323-2324]), Σ_0 is uniquely homologically area-minimising in a tubular neighbourhood. Applying [MN26, Corollary 1] we find a metric h_0 on M for which Σ_0 bounds an isoperimetric region in (M, h_0) with $h_0 = g_0$ in a tubular neighbourhood $V_0 \subset M$ of Σ_0 (in which Σ_0 remains locally uniquely homologically area-minimising).

We now consider a sequence $H_j \to 0$ (with $H_j \neq 0$) and argue that for $j \geq 1$ sufficiently large we can find H_j-CMC hypersurfaces $\Sigma_j = \partial\Omega_j$ with Ω_j an isoperimetric region for its volume with respect to some Riemannian metric h_j. Along this sequence, Proposition 1 (now applied in (M, h_0)) produces separating, strictly stable H_j-CMC hypersurfaces $\Sigma_j = \partial\Omega_j$ for some Riemannian metric h_j, where $h_j = h_0$ outside of annular regions around $\mathrm{Sing}(\Sigma_0)$ that are contained in V_0. Moreover, from its construction (up to choosing larger $j \geq 1$) we can arrange that $\Sigma_j \subset V_0$, each Σ_j is homologous to Σ_0 in V_0, with $\Sigma_j \to \Sigma_0$ in Hausdorff distance, $h_j \to h$ in C^3, and $\mathrm{Per}_{h_j}(\Omega_j) \to \mathrm{Per}_{h_0}(\Omega_0)$ as $j \to \infty$.

For a sequence $t_k \to 0$ we consider the barriers S_k^j (for fixed $j \geq 1$) constructed over each Σ_j at the start of Subsection 2.3. Applying Theorem 4, we deduce that each Ω_j is uniquely $\mathcal{F}_{H_j}$-minimising with respect to the metric h_j in S_k^j for all $k \geq 1$ sufficiently large; let us denote $k_j \geq 1$ for the smallest such $k \geq 1$ for which this holds (noting that $S_{k+1}^j \subset S_k^j$). We now show that $\tilde{k} = \sup_{j \geq 1} k_j$ is finite, demonstrating that the tubular neighbourhoods of the Σ_j in which the Ω_j are uniquely volume-constrained minimising are uniformly bounded below in size. If this supremum was not finite we would find a sequence $k_j \to \infty$ and $\mathcal{F}_{H_j}$-minimisers $E_j \neq \Omega_j$ in $S_{k_j-1}^j$ for each $j \geq 1$. Since $t_{k_j} \to 0$ we have $\partial S_{k_j-1}^j \to \Sigma_0$ and hence $E_j \to \Omega_0$ as $j \to \infty$. Arguing as in the proof of Theorem 4, now over

each hypersurface Σ_j, we deduce that $E_j = \Omega_j$ for sufficiently large $j \geq 1$ which is a contradiction. Here we are using the fact that the strict stability constants of the Σ_j are uniformly bounded below, that the constants in Lemma 1 and Remark 3 do not depend on $j \geq 1$, and that each Σ_j has isolated singularities modelled on the same tangent cones $\{\mathbf{C}_i\}_{i=1}^L$ as Σ_0.

From the construction of the Σ_j over Σ_0 we ensure that for fixed $k \geq 1$ the barriers S_k^j have a width bounded below independently of $j \geq 1$ and that $\Sigma_j \to \Sigma_0$ in the Hausdorff sense as $j \to \infty$. We thus see that S_k^j contains a tubular neighbourhood V (with smooth boundary) of Σ_0 for all $j \geq 1$ sufficiently large with $\Sigma_j \subset V \subset S_k^j$ for all $j \geq 1$ sufficiently large. Moreover, from the above we ensure that Ω_j is uniquely $\mathcal{F}_{H_j}$-minimising in V whenever $j \geq 1$ is sufficiently large.

If $h_j = h_0$ we could directly apply [MN26, Corollary 2] in V (noting that by [MN26, Remark 2] we allow for non-empty singular set) to conclude that Ω_j is an isoperimetric region with respect to the metric h_j whenever $j \geq 1$ was sufficiently large. However, since we do not necessarily have $h_j = h_0$, we must adapt its proof; we now outline this and refer the reader to the proof of [MN26, Corollary 2].

First, we note that the construction of h_0 (which agrees with h_j outside of V) from the proof of [MN26, Theorem 1] ensures that any isoperimetric region E_j of enclosed volume $\mathrm{Vol}_{h_j}(\Omega_j)$ with respect to h_j must have some boundary component homologous to Σ_0 (and hence Σ_j) in V_0. Since $\mathrm{Per}_{h_j}(\Omega_j) \to \mathrm{Per}_{h_0}(\Omega_0)$ as $j \to \infty$ and Σ_0 is uniquely homologically area-minimising in V with respect to h_0, potentially taking $j \geq 1$ larger we ensure that E_j has only one boundary component which moreover must lie in V since $\partial E_j \to \Sigma_0$ as $j \to \infty$). Indeed, this holds whenever $|\mathrm{Per}_{h_j}(\Omega_j) - \mathrm{Per}_{h_0}(\Omega_0)| < \frac{\delta}{2}$, where $\delta > 0$ is a uniform lower bound for the perimeter of each boundary component of this isoperimetric region. Such a δ arises from the uniformly bounded geometry of the h_j (noting again that $h_j \to h_0$ as $j \to \infty$), which gives a uniform mean curvature bound for ∂E_j, from which an application of the monotonicity formula as in the proof of [MN26, Theorem 1].

Thus we see that for $j \geq 1$ sufficiently large, any isoperimetric region E_j of enclosed volume $\mathrm{Vol}_{h_j}(\Omega_j)$ with respect to h_j has only one boundary component, with this component lying in V, and homologous to Σ_j. The unique volume-constrained minimisation of Ω_j in V then ensures that $E_j = \Omega_j$, so that Ω_j is an isoperimetric region for its volume with a singular non-zero CMC hypersurface Σ_j as its boundary. □

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Declaration

The ideas presented in this work were developed by the authors.

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